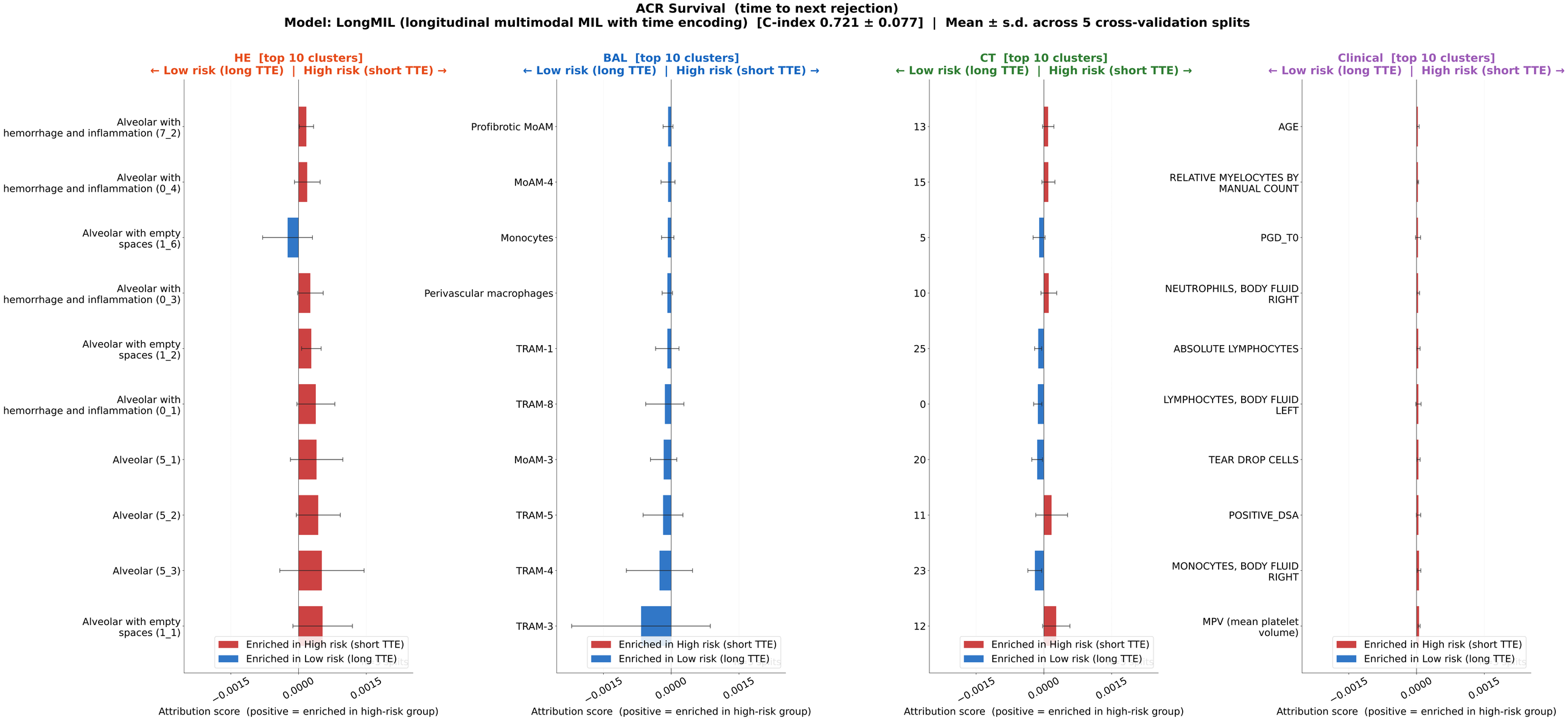
ACR Survival (time to next rejection)
Model: LongMIL (longitudinal multimodal MIL with time encoding) [C-index 0.721 ± 0.077] | Mean ± s.d. across 5 cross-validation splits
HE [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
Alveolar with hemorrhage and inflammation (7_2)
Alveolar with hemorrhage and inflammation (0_4)
Alveolar with empty spaces (1_6)
Alveolar with hemorrhage and inflammation (0_3)
Alveolar with empty spaces (1_2)
Alveolar with hemorrhage and inflammation (0_1)
Alveolar (5_1)
Alveolar (5_2)
Alveolar (5_3)
Alveolar with empty spaces (1_1)
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0015
0.0000
0.0015
Attribution score (positive = enriched in high-risk group)
BAL [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
Profibrotic MoAM
MoAM-4
Monocytes
Perivascular macrophages
TRAM-1
TRAM-8
MoAM-3
TRAM-5
TRAM-4
TRAM-3
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0015
0.0000
0.0015
Attribution score (positive = enriched in high-risk group)
CT [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
13
15
5
10
25
0
20
11
23
12
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0015
0.0000
0.0015
Attribution score (positive = enriched in high-risk group)
Clinical [top 10 clusters]
← Low risk (long TTE) | High risk (short TTE) →
AGE
RELATIVE MYELOCYTES BY MANUAL COUNT
PGD_T0
NEUTROPHILS, BODY FLUID RIGHT
ABSOLUTE LYMPHOCYTES
LYMPHOCYTES, BODY FLUID LEFT
TEAR DROP CELLS
POSITIVE_DSA
MONOCYTES, BODY FLUID RIGHT
MPV (mean platelet volume)
Enriched in High risk (short TTE)
Enriched in Low risk (long TTE)
−0.0015
0.0000
0.0015
Attribution score (positive = enriched in high-risk group)